

4.4 Surface Area; Parametric Surfaces

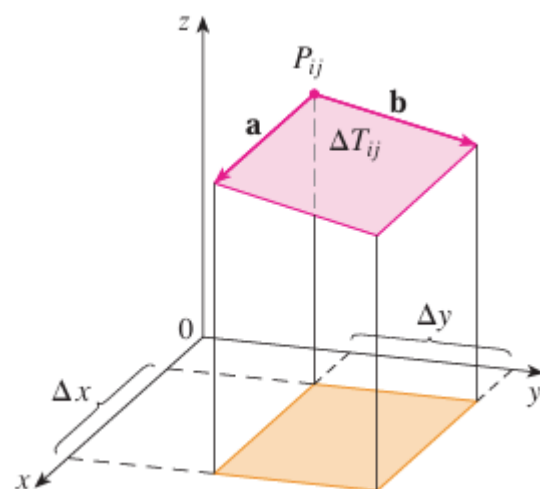
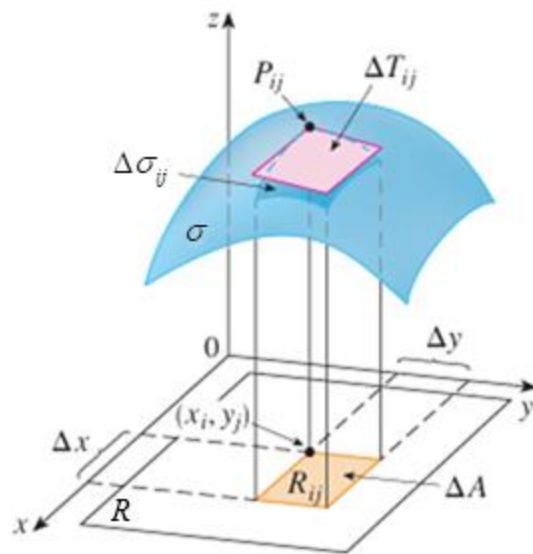
Motivation: Suppose that σ is a surface with equation $z = f(x, y)$ where f has continuous partial derivatives. Suppose also that $f(x, y) \geq 0$, and we wish to find the surface area of σ that exists over some region R in the xy -plane.

Our strategy for finding the surface area will be to divide R up into small rectangles R_{ij} with area $\Delta A = \Delta x \Delta y$ and thereby split up σ into smaller “patches” which we will call $\Delta\sigma_{ij}$. We then want to approximate the area of each $\Delta\sigma_{ij}$ over the rectangle R_{ij} . A picture of what a $\Delta\sigma_{ij}$ over R_{ij} might look like is shown at the right.

In actuality there would be many “pieces” $\Delta\sigma_{ij}$ of σ like the one pictured because there would be many rectangles R_{ij} . We are just focusing on one of the pieces.

Our plan is to now consider the point $P_{ij}(x_i, y_j)$ in R_{ij} closest to the origin, then $(x_i, y_j, f(x_i, y_j))$ is a point on the surface directly above the corner of R_{ij} closest to the origin. At this point we will construct a tangent plane to the surface (shown in pink above) and we will use the area of the tangent plane ΔT_{ij} (which is just the area of a parallelogram) to estimate the area of the surface $\Delta\sigma_{ij}$.

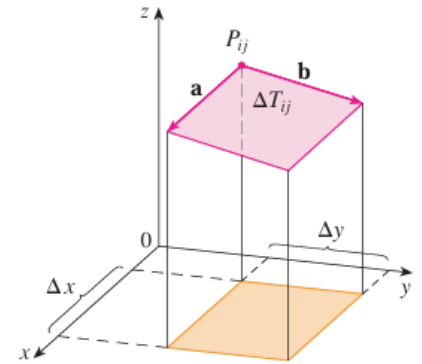
Now suppose we build vectors $\mathbf{a}$ and $\mathbf{b}$ that start at P_{ij} and lie along the sides of the parallelogram (as shown in the image at the right). We let ΔT_{ij} represent the area of this parallelogram. We note that as R_{ij} gets smaller and smaller ΔT_{ij} will do a better job of approximating the “piece” of σ . This leads to the following definition.



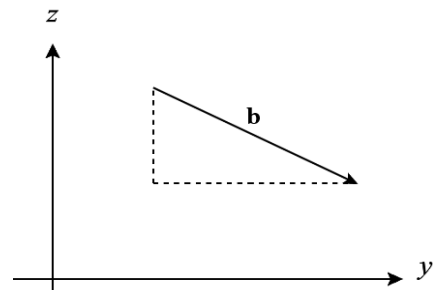
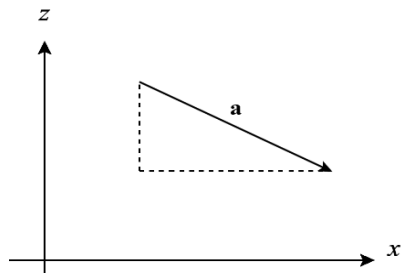
Definition: The surface area of σ (as described above) is

Finding ΔT_{ij} (the area of the portion of the tangent plane we desire) can be done easily using ideas from Chapter 1. Recall that in order to find the area of a parallelogram determined by vectors **a** and **b** we simply compute

To complete this computation, we need to find the components of **a** and **b**. From the picture we see that



To find the z components we note that both **a** and **b** share the same direction (3D “slope”) as the tangent lines through the point P_{ij} . Based on our knowledge from Chapter 3, we know these tangent lines have slopes of $f_x(x_i, y_j)$ & $f_y(x_i, y_j)$.



and therefore

Therefore,

Now we can compute the area of the parallelogram determined by vectors **a** and **b**.

This, along with a generalization that proves this idea works for surfaces over regions besides rectangles, leads to the following theorem.

Theorem: The area of the surface with equation $z = f(x, y)$ where $(x, y) \in D$ and the first partials of f are continuous is:

$$\begin{aligned} S &= \iint_D \left(\sqrt{(f_x(x, y))^2 + (f_y(x, y))^2 + 1} \right) dA \\ &= \iint_D \left(\sqrt{\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 + 1} \right) dA. \end{aligned}$$

Example 1: Set up but do not evaluate an iterated integral which yields the surface area of the surface determined by $z = 1 + 2x + 3y + 4y^2$ that lies above the triangle with vertices $(0, 0)$, $(1, 1)$ and $(2, 1)$.

Example 2: Find the exact surface area of the part of the surface given by $z = 4 - x^2 - y^2$ that lies above the xy -plane.

Parametric Surfaces and Their Areas

Recall:

In chapter 2 we learned that vector functions of the form $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^3$ is a powerful way to represent curves in the three space. We also learned that vector functions of this form could do a very nice job modeling the movement of a particle in three dimensions.

Then, in chapter 3 we learned that the graph of a function of two variables can be graphed as a surface in three dimensions.

In this section we generalize the ideas presented in chapter 2 to surfaces. This leads to a more powerful way of representing surfaces (than was learned in chapter 3).

Definition: The set of all points in $\mathbb{R}^3$ such that $x = x(u, v)$ $y = y(u, v)$ $z = z(u, v)$ where (u, v) varies throughout some subset, D , of $\mathbb{R}^2$, is called a parametric surface σ and the three equations are called the parametric equations of σ .

Note: Each choice of (u, v) gives a point on σ ; by making all possible choices we get all of σ . A nice illustration of how this works can be found in figure 14.4.5 on page 1028 of the book.

Note: We can represent a parametric surface by a vector function of the form

$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$ *where we obtain a picture of the surface by thinking of each output of $\mathbf{r}$ as a position vector and we graph only its terminal point.*

Example 3: Identify and sketch the surface given by the vector function

$$\mathbf{r}(u, v) = (2\cos(u))\mathbf{i} + (v)\mathbf{j} + (2\sin(u))\mathbf{k}.$$

Example 4: Find parametric equations of the plane given by $x + y + z = 1$.

Example 5:

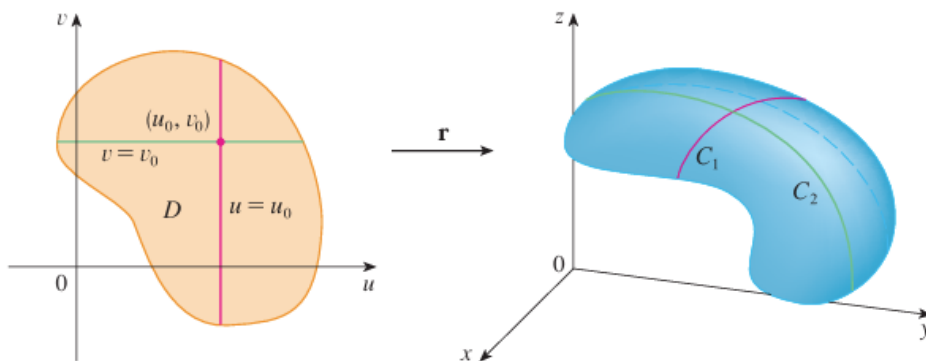
a) What type of quadric surface does $\mathbf{r}(u, v) = \langle u, v, v^2 - u^2 \rangle$ yield?

b) Find a vector valued function $\mathbf{r}(u, v)$ which yields the elliptical cylinder given by: $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

Exploration:

Now, suppose that we would like to find the tangent plane to the surface σ (at some point on σ) where σ is traced out by a vector function $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$.

Specifically suppose we want the tangent plane to the surface at some point P_0 with position vector $\mathbf{r}(u_0, v_0)$. Suppose we consider the grid curve on σ formed by $\mathbf{r}(u, v)$ when we let $u = u_0$ and the grid curve on σ formed by $\mathbf{r}(u, v)$ when we let $v = v_0$. A picture of the situation is:



The curve on σ in red is the grid curve formed by

The curve on σ in green is the grid curve formed by

We want the tangent plane to contain both the vectors

So let $\mathbf{n} =$

Definition: Suppose a surface σ is given by $\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$. If $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$ then the surface σ is called smooth.

Note: Intuitively, a surface is smooth if it has no sharp corners. We can also say that a surface, σ , is smooth if we can find a tangent plane to σ at every point on σ .

Fact: Suppose a surface σ is given by $\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$. If $\mathbf{r}(u_0, v_0) = \langle x_0, y_0, z_0 \rangle$ and $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0) \neq \mathbf{0}$, then an equation of the tangent plane to σ at the point (x_0, y_0, z_0) is

Example 6: Find an equation of the tangent plane to the surface given by $\mathbf{r}(u,v) = \langle u^2, v^2, u + 2v \rangle$ at the point $(1,1,3)$.

As we did previously we can approximate the surface area of a surface by using portions of tangent planes, and as our partition gets finer our approximation approaches perfection.

This idea leads to the following definition.

Definition: If a smooth parametric surface σ is given by the equation $\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$ where $(u,v) \in D$ and σ is covered just once as (u,v) ranges throughout the parameter domain D , then the surface area of σ is $S = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| dA$.

Note: See page 1034-1035 of the book for the Riemann sum version of the above definition.

Example 7: Find the exact surface area of the portion of the spiral ramp given by

$$\mathbf{r}(u, v) = \langle u \cos(v), u \sin(v), v \rangle \quad \text{where} \quad (u, v) \in \{(u, v) : 0 \leq u \leq 1, 0 \leq v \leq u\}$$

Remark: Suppose we want to find the surface area of the surface determined by $z = f(x, y)$ where $(x, y) \in D$. Rather than using the ideas from the beginning of this section, we can do this with the theory developed about parametric surfaces.